CHAPTER III

Figure and Ground

Primes *vs.* Composites

THERE IS A strangeness to the idea that concepts can be captured by simple typographical manipulations. The one concept so far captured is that of addition, and it may not have appeared very strange. But suppose the goal were to create a formal system with theorems of the form **P**x, the letter `x' standing for a hyphen-string, and where the only such theorems would be ones in which the hyphen-string contained exactly a prime number of hyphens. Thus**, P---** would be a theorem, but **P----** would not. How could this be done typographically? First, it is important to specify clearly what is meant by *typographical* operations. The complete repertoire has been presented in the **MIU**-system and the pq-system, so we really only need to make a list of the kinds of things we have permitted:

(1) reading and recognizing any of a finite set of symbols;
(2) writing down any symbol belonging to that set;
(3) copying any of those symbols from one place to another;
(4) erasing any of those symbols;
(5) checking to see whether one symbol is the same as another;
(6) keeping and using a list of previously generated theorems.

The list is a little redundant, but no matter. What is important is that it clearly involves only trivial abilities, each of them far less than the ability to distinguish primes from nonprimes. How, then, could we compound some of these operations to make a formal system in which primes are distinguished from composite numbers?

The tq-System

A first step might be to try to solve a simpler, but related, problem. We could try to make a system similar to the pq-system, except that it represents multiplication, instead of addition. Let's call it the *tq-system*, `t' for times'. More specifically, suppose X, Y, and Z are, respectively, the numbers of hyphens in the hyphen-strings x, y, and z. (Notice I am taking special pains to distinguish between a string and the number of hyphens it contains.) Then we wish the string x ty q z to be a theorem if and only if X times Y equals Z. For instance, **--t---q-----** should be a theorem because 2 times 3 equals 6, but --t--q--- should not be a theorem. The tq-system can be characterized just about as easily as the pq-system namely, by using just one axiom schema and one rule of inference:

AXIOM SCHEMA: x**t**-**q**x is an axiom, whenever x is a hyphen string.

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RULE OF INFERENCE: Suppose that x, y, and z are all hyphen-strings. An suppose that x **t**y **q**z is an old theorem. Then, xty-qzx is a ne' theorem.

Below is the derivation of the theorem **--t---q-----**

:

(1)	--t-q--	(axiom)
(2)	--t—q----	(by rule of inference, using line (1) as the old theorem)
(3)	--t---q -------	(by rule of inference, using line (2) as the old theorem)

Notice how the middle hyphen-string grows by one hyphen each time the rule of inference is applied; so it is predictable that if you want a theorem with ten hyphens in the middle, you apply the rule of inference nine times in a row.

Capturing Compositeness

Multiplication, a slightly trickier concept than addition, has now bee] "captured" typographically, like the birds in Escher's Liberation. What about primeness? Here's a plan that might seem smart: using the tq-system define a new set of theorems of the form **C**x, which characterize compost. numbers, as follows:

RULE: Suppose x, y, and z are hyphen-strings. If x-ty-qz is a theorem then **C** z is a theorem.

This works by saying that Z (the number of hyphens in z) is composite a long as it is the product of two numbers greater than 1-namely, X + (the number of hyphens in x-), and Y + 1 (the number of hyphens in y I am defending this new rule by giving you some "Intelligent mode justifications for it. That is because you are a human being, and want t, know *why* there is such a rule. If you were operating exclusively in the "Mechanical mode", you would not need any justification, since M-mod. workers just follow the rules mechanically and happily, never questioning; them!

Because you work in the I-mode, you will tend to blur in your mind the distinction between strings and their interpretations. You see, things Cal become quite confusing as soon as you perceive "meaning" in the symbol which you are manipulating. You have to fight your own self to keep from thinking that the *string'*---' is the *number* 3. The Requirement of Formality, which in Chapter I probably seemed puzzling (because it seemed so obvious), here becomes tricky, and crucial. It is the essential thing which keeps you from mixing up the I-mode with the M-mode; or said another way, it keeps you from mixing up arithmetical facts with typographical theorems.

Illegally Characterizing Primes

It is very tempting to jump from the C-type theorems directly to P-type theorems, by proposing a rule of the following kind:

PROPOSED RULE: Suppose x is a hyphen-string. If **C**x is not a theorem, then **P**x is a theorem.

The fatal flaw here is that checking whether **C**x is not a theorem is not an explicitly typographical operation. To know for sure that **MU** is not a theorem of the MIU-system, you have to go *outside* of the system ... and so it is with this Proposed Rule. It is a rule which violates the whole idea of formal systems, in that it asks you to operate informally-that is, outside the system. Typographical operation (6) allows you to look into the stockpile of previously found theorems, but this Proposed Rule is asking you to look into a hypothetical "Table of Nontheorems". But in order to generate such a table, you would have to do some reasoning *outside the system*-reasoning which shows why various strings cannot be generated inside the system. Now it may well be that there is another formal system which can generate the "Table of Nontheorems", by purely typographical means. In fact, our aim is to find just such a system. But the Proposed Rule is not a typographical rule, and must be dropped.

This is such an important point that we might dwell on it a bit more. In our **C**-system (which includes the tq-system and the rule which defines C-type theorems), we have theorems of the form **C**x, with `x' standing, as usual, for a hyphen-string. There are also nontheorems of the form **C**x. (These are what I mean when I refer to "nontheorems", although of course **tt-Cqq** and other ill-formed messes are also nontheorems.) The difference is that theorems have a composite number of hyphens, nontheorems have a prime number of hyphens. Now the theorems all have a common "form", that is, originate from a common set of typographical rules. Do all nontheorems also have a common "form", in the same sense? Below is a list of C-type theorems, shown without their derivations. The parenthesized numbers following them simply count the hyphens in them.

C---- (4)
C-------- (6)
C----------(8)
C----------(9)
C----------(10)
C ---------- (12)
C---------- (14)
C ---------- (15)
C----------(16)
C----------(18)

•
•
•

I he "holes" in this list are the nontheorems. I o repeat the earlier quest Do the holes also have some "form" in common? Would it be reasonable say that merely by virtue of being the holes in this list, they share a common form? Yes and no. That they share *some* typographical quality is and able, but whether we want to call it "form" is unclear. The reason hesitating is that the holes are only *negatively* defined-they are the things that are left out of a list which is *positively* defined.

Figure and Ground

This recalls the famous artistic distinction between *figure* and *ground*. When a figure or "positive space" (e.g., a human form, or a letter, or a still life is drawn inside a frame, an unavoidable consequence is that its complementary shape-also called the "ground", or "background", or "negative space"-has also been drawn. In most drawings, however, this fig ground relationship plays little role. The artist is much less interested in ground than in the figure. But sometimes, an artist will take interest in ground as well.

There are beautiful alphabets which play with this figure-ground distinction. A message written in such an alphabet is shown below. At fir looks like a collection of somewhat random blobs, but if you step back ways and stare at it for a while, all of a sudden, you will see seven letters appear in this ..

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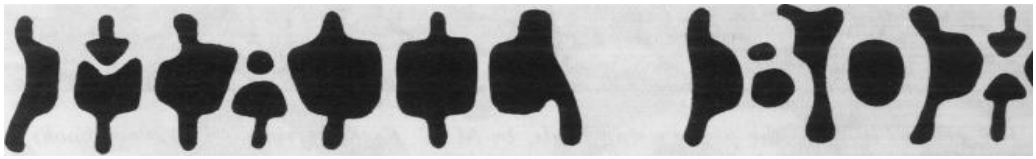

FIGURE 15.

For a similar effect, take a look at my drawing *Smoke Signal* (Fig. 139). Along these lines, you might consider this puzzle: can you somehow create a drawing containing words in both the figure *and* the ground?

Let us now officially distinguish between two kinds of figures: *cursively drawable* ones, and *recursive* ones (by the way, these are my own terms are not in common usage). A *cursively drawable* figure is one whose ground is merely an accidental by-product of the drawing act. A *recursive* figure is one whose ground can be seen as a figure in its own right. Usually this is quite deliberate on the part of the artist. The "re" in "recursive" represents the fact that both foreground and background are cursively drawable – the figure is "twice-cursive". Each figure-ground boundary in a recursive figure is a double-edged sword. M. C. Escher was a master at drawing recursive figures-see, for instance, his beautiful recursive drawing of birds (Fig. 16).

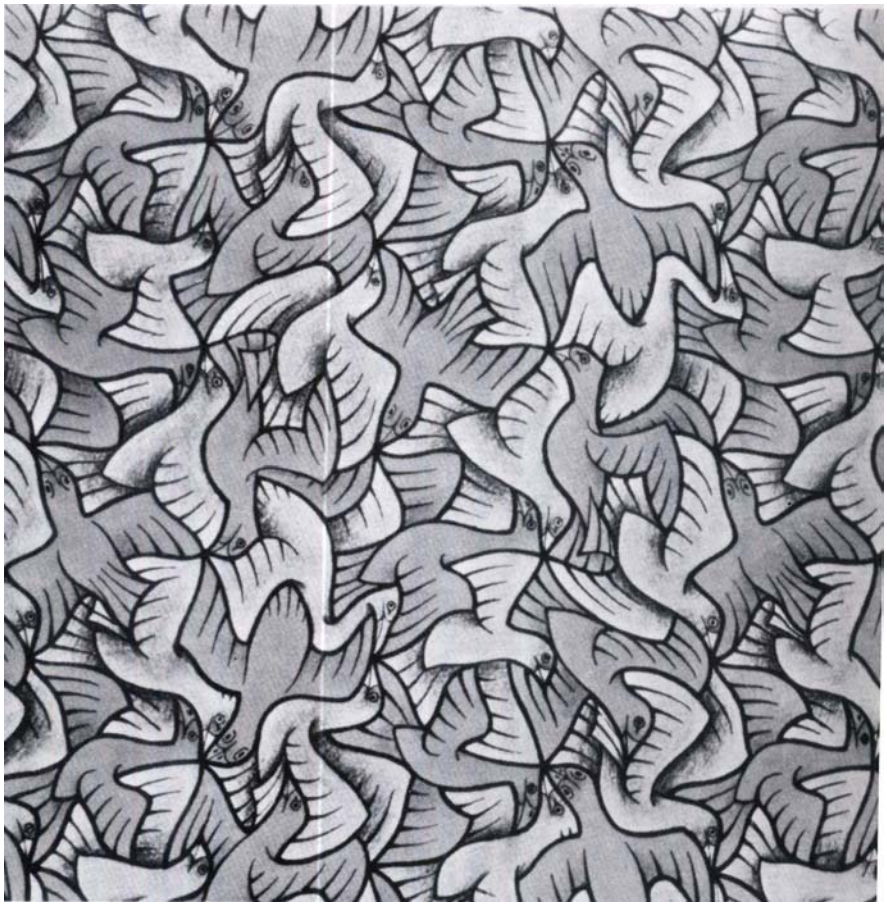

FIGURE 16. Tiling of the plane using birds, by M. C. Escher (from a 1942 notebook).

Our distinction is not as rigorous as one in mathematics, for who can definitively say that a particular ground is not a figure? Once pointed out, almost any ground has interest of its own. In that sense, every figure is recursive. But that is not what I intended by the term. There is a natural and intuitive notion of recognizable forms. Are both the foreground and background recognizable forms? If so, then the drawing is recursive. If you look at the grounds of most line drawings, you will find them rather unrecognizable. This demonstrates that

There exist recognizable forms whose negative space is not any recognizable form.

In more "technical" terminology, this becomes:

There exist cursively drawable figures which are not recursive.

Scott Kim's solution to the above puzzle, which I call his "FIGURE-FIGURE Figure", is shown in Figure 17. If you read both black and white,

FIGURE 17. FIGURE-FIGURE Figure, by Scott E. Kim (1975).

you will see "FIGURE" everywhere, but "GROUND" nowhere! It is a paragon of recursive figures. In this clever drawing, there are two nonequivalent ways of characterizing the black regions:

(1) as the *negative space* to the white regions;
(2) as *altered copies* of the white regions (produced by coloring and shifting each white region).

(In the special case of the FIGURE-FIGURE Figure, the two characterizations are equivalent-but in most black-and-white pictures, they would not be.) Now in Chapter VIII, when we create our Typographical Number Theory (**TNT**), it will be our hope that the set of all false statements of number theory can be characterized in two analogous ways:

(1) as the negative space to the set of all **TNT**-theorems;
(2) as altered copies of the set of all **TNT**-theorems (produced by negating each **TNT**-theorem).

But this hope will be dashed, because:

(1) inside the set of all nontheorems are found some truths
(2) outside the set of all negated theorems are found some falsehoods
.

You will see why and how this happens, in Chapter XIV. Meanwhile, ponder over a pictorial representation of the situation (Fig. 18).

Figure and Ground in Music

One may also look for figures and grounds in music. One analogue is the distinction between melody and accompaniment-for the melody is always in the forefront of our attention, and the accompaniment is subsidiary, in some sense. Therefore it is surprising when we find, in the lower lines of a piece of music, recognizable melodies. This does not happen too often in post-baroque music. Usually the harmonies are not thought of as foreground. But in baroque music-in Bach above all-the distinct lines, whether high or low or in between, all act as "figures". In this sense, pieces by Bach can be called "recursive".

Another figure-ground distinction exists in music: that between on-beat and off-beat. If you count notes in a measure "one-and, two-and, three-and, four-and", most melody-notes will come on numbers, not on "and"'s. But sometimes, a melody will be deliberately pushed onto the "and" 's, for the sheer effect of it. This occurs in several etudes for the piano by Chopin, for instance. It also occurs in Bach-particularly in his Sonatas and Partitas for unaccompanied violin, and his Suites for unaccompanied cello. There, Bach manages to get two or more musical lines going simultaneously. Sometimes he does this by having the solo instrument play "double stops"-two notes at once. Other times, however, he

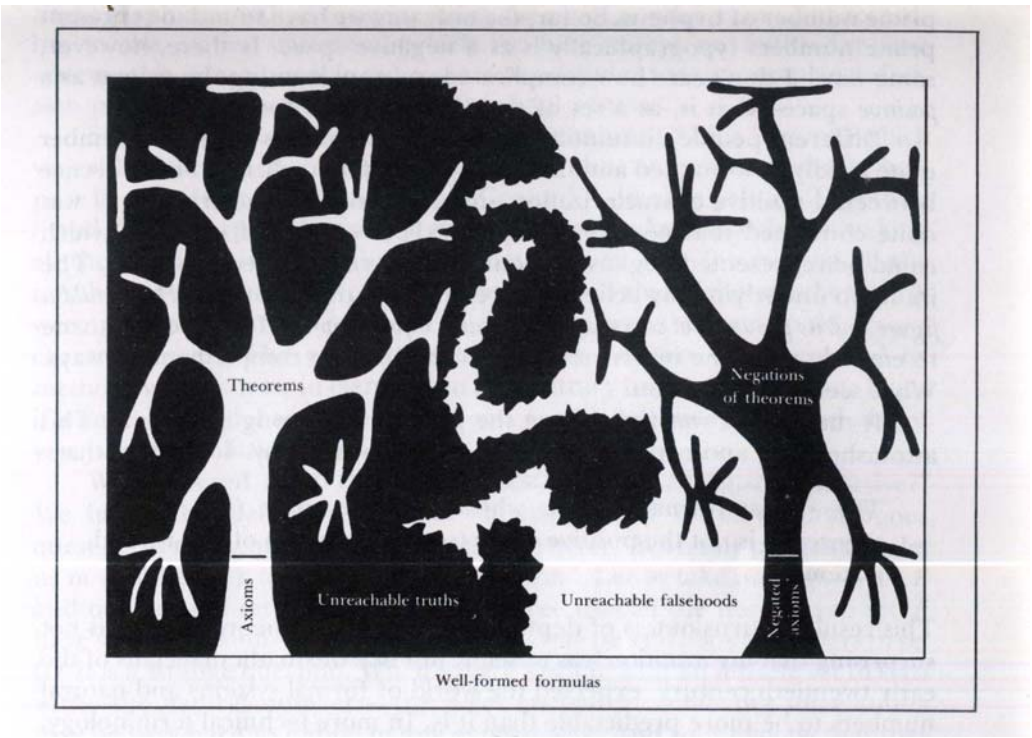

*FIGURE 18. Considerable visual symbolism is featured in this diagram of the relation between various classes of **TNT** strings. The biggest box represents the set of all **TNT** strings The next-biggest box represents the set of all well-formed TNT strings. Within it is found~ set of all sentences of **TNT**. Now things begin to get interesting. The set of theorems pictured as a tree growing out of a trunk (representing the set of axioms). The tree-symbol chosen because of the recursive growth pattern which it exhibits: new branches (theorems constantly sprouting from old ones. The fingerlike branches probe into the corners of constraining region (the set of truths), yet can never fully occupy it. The boundary beta the set of truths and the set of falsities is meant to suggest a randomly meandering coastline which, no matter how closely you examine it, always has finer levels of structure, an consequently impossible to describe exactly in any finite way. (See B. Mandelbrot's book Fractals.) The reflected tree represents the set of negations of theorems: all of them false yet unable collectively to span the space of false statements. [Drawing by the author.]*

puts one voice on the on-beats, and the other voice on the off-beats, so ear separates them and hears two distinct melodies weaving in and out, - harmonizing with each other. Needless to say, Bach didn't stop at this level of complexity...

Recursively Enumerable Sets vs. Recursive Sets

Now let us carry back the notions of figure and ground to the domain formal systems. In our example, the role of positive space is played by **C**-type theorems, and the role of negative space is played by strings with a

prime number of hyphens. So far, the only way we have found to represent prime numbers typographically is as a negative space. Is there, however, some way-I don't care how complicated-of representing the primes as a *positive* space-that is, as a set of theorems of some formal system?

Different people's intuitions give different answers here. I remember quite vividly how puzzled and intrigued I was upon realizing the difference between a positive characterization and a negative characterization. I was quite convinced that not only the primes, but any set of numbers which could be represented negatively, could also be represented positively. The intuition underlying my belief is represented by the question: "*How could a figure and its ground not carry exactly the same information*?" They seemed to me to embody the same information, just coded in two complementary ways. What seems right to you?

It turns out I was right about the primes, but wrong in general. This astonished me, and continues to astonish me even today. It is a fact that:

> There exist formal systems whose negative space (set of nontheorems) is not the positive space (set of theorems) of any formal system.

This result, it turns out, is of depth equal to Gödel's Theorem-so it is not surprising that my intuition was upset. I, just like the mathematicians of the early twentieth century, expected the world of formal systems and natural numbers to be more predictable than it is. In more technical terminology, this becomes:

> There exist recursively enumerable sets which are not recursive.

The phrase *recursively enumerable* (often abbreviated "r.e.") is the mathematical counterpart to our artistic notion of "cursively drawable"-and *recursive* is the counterpart of "recursive". For a set of strings to be "r.e." means that it *can* be generated according to typographical rules-for example, the set of **C**-type theorems, the set of theorems of the **MIU**-system-indeed, the set of theorems of any formal system. This could be compared with the conception of a "figure" as "a set of lines which can be generated according to artistic rules" (whatever that might mean!). And a "recursive set" is like a figure whose ground is also a figure-not only is it r.e., but its complement is also r.e.

It follows from the above result that:

> There exist formal systems for which there is no typographical decision procedure.

How does this follow? Very simply. A typographical decision procedure is a method which tells theorems from nontheorems. The existence of such a test allows us to generate all nontheorems systematically, simply by going down a list of *all* strings and performing the test on them one at a time, discarding ill-formed strings and theorems along the way. This amounts to

a typographical method for generating the set of nontheorems. But according to the earlier statement (which we here accept on faith), for *some* systems this is not possible. So we must conclude that typographical decision procedures do not exist for all formal systems.

Suppose we found a set F of natural numbers (`*F*' for `Figure') whi4 we could generate in some formal way-like the composite numbers. Suppose its complement is the set G (for 'Ground')-like the primes. Together F and G make up all the natural numbers, and we know a rule for making all the numbers in set F, but we know no such rule for making all tl numbers in set G. It is important to understand that if the members of were always generated in order of *increasing size*, then we could always characterize G. The problem is that many r.e. sets are generated I methods which throw in elements in an arbitrary order, so you never know if a number which has been skipped over for a long time will get included you just wait a little longer.

We answered no to the artistic question, "Are all figures recursive We have now seen that we must likewise answer no to the analogous question in mathematics: "Are all sets recursive?" With this perspective, 1 us now come back to the elusive word "form". Let us take our figure-set and our ground-set G again. We can agree that all the numbers in set have some common "form"-but can the same be said about numbers in s G? It is a strange question. When we are dealing with an infinite set to sta with-the natural numbers-the holes created by removing some subs may be very hard to define in any explicit way. And so it may be that th< are not connected by any common attribute or "form". In the last analysis it is a matter of taste whether you want to use the word "form"-but just thinking about it is provocative. Perhaps it is best not to define "form", bi to leave it with some intuitive fluidity.

Here is a puzzle to think about in connection with the above matter Can you characterize the following set of integers (or its negative space)

1 3 7 12 18 26 35 45 56 69...

How is this sequence like the FIGURE-FIGURE Figure?

Primes as Figure Rather than Ground

Finally, what about a formal system for generating primes? How is it don< The trick is to skip right over multiplication, and to go directly to *nondivisibility* as the thing to represent positively. Here are an axiom schema and rule for producing theorems which represent the notion that one number does not divide (**D N D**) another number exactly:

AXIOM SCHEMA: xy **D N D**x where x and y are hyphen-strings.

For example **----D N D--,** where x has been replaced by'--'and y by '---".

RULE: If x **D N D**y is a theorem, then so is x **D N D**x y.

If you use the rule twice, you can generate this theorem:

-----D N D ----------

which is interpreted as "5 does not divide 12". But **---D N D----------** is not a theorem. What goes wrong if you try to produce it?

Now in order to determine that a given number is prime, we have to build up some knowledge about its nondivisibility properties. In particular, we want to know that it is not divisible by 2 or 3 or 4, etc., all the way up to 1 less than the number itself. But we can't be so vague in formal systems as to say "et cetera". We must spell things out. We would like to have a way of saying, in the language of the system, "the number Z is *divisor free* up to X", meaning that no number between 2 and X divides Z. This can be done, but there is a trick to it. Think about it if you want.

Here is the solution:

RULE: If **--D N D** z is a theorem, so is z **D F**--.

RULE: If z **D F**x is a theorem and also x-**D N D**z is a theorem, z **D F**x- is a theorem.

These two rules capture the notion of divisor freeness. All we need to do is to say that primes are numbers which are divisor-free up to 1 less than themselves:

RULE: If z-**DF**z is a theorem, then **P**z- is a theorem.

Oh-let's not forget that 2 is prime!

Axiom: **P--.**

And there you have it. The principle of representing primality formally is that there is a test for divisibility which can be done without any backtracking. You march steadily upward, testing first for divisibility by 2, then by 3, and so on. It is this "monotonicity" or unidirectionality-this absence of cross-play between lengthening and shortening, increasing and decreasing-that allows primality to be captured. And it is this potential complexity of formal systems to involve arbitrary amounts of backwards-forwards interference that is responsible for such limitative results as Gödel's Theorem, Turing's Halting Problem, and the fact that not all recursively enumerable sets are recursive.

Contracrostipunctus

Achilles has come to visit his friend and jogging companion, the Tortoise, at his home

Achilles: Heavens, you certainly have an admirable boomerang collection

Tortoise: Oh, pshaw. No better than that of any other Tortoise. And now would you like to step into the parlor?

Achilles: Fine. (*Walks to the corner of the room.*) I see you also have a large collection of records. What sort of music do you enjoy?

Tortoise: Sebastian Bach isn't so bad, in my opinion. But these days, I must say, I am developing more and more of an interest in a rather specialized sort of music.

Achilles: Tell me, what kind of music is that?

Tortoise: A type of music which you are most unlikely to have heard of. call it "music to break phonographs by".

Achilles: Did you say "to break phonographs by"? That is a curious concept. I can just see you, sledgehammer in hand, whacking on phonograph after another to pieces, to the strains of Beethoven's heroic masterpiece *Wellington's Victory*.

Tortoise: That's not quite what this music is about. However, you might find its true nature just as intriguing. Perhaps I should give you a brief description of it?

Achilles: Exactly what I was thinking.

Tortoise: Relatively few people are acquainted with it. It all began whet my friend the Crab-have you met him, by the way?-paid m• a visit.

Achilles: ' twould be a pleasure to make his acquaintance, I'm sure Though I've heard so much about him, I've never met him

Tortoise: Sooner or later I'll get the two of you together. You'd hit it of splendidly. Perhaps we could meet at random in the park on day ...

Achilles: Capital suggestion! I'll be looking forward to it. But you were going to tell me about your weird "music to smash phone graphs by", weren't you?

Tortoise: Oh, yes. Well, you see, the Crab came over to visit one day. You must understand that he's always had a weakness for fang gadgets, and at that time he was quite an aficionado for, of al things, record players. He had just bought his first record player, and being somewhat gullible, believed every word the salesman had told him about it-in particular, that it was capable of reproducing any and all sounds. In short, he was convinced that it was a Perfect phonograph.

Achilles: Naturally, I suppose you disagreed.

Tortoise: True, but he would hear nothing of my arguments. He staunchly maintained that any sound whatever was reproducible on his machine. Since I couldn't convince him of the contrary, I left it at that. But not long after that, I returned the visit, taking with me a record of a song which I had myself composed. The song was called "I Cannot Be Played on Record Player 1".

Achilles: Rather unusual. Was it a present for the Crab?

Tortoise: Absolutely. I suggested that we listen to it on his new phonograph, and he was very glad to oblige me. So he put it on. But unfortunately, after only a few notes, the record player began vibrating rather severely, and then with a loud "pop", broke into a large number of fairly small pieces, scattered all about the room. The record was utterly destroyed also, needless to say.

Achilles: Calamitous blow for the poor fellow, I'd say. What was the matter with his record player?

Tortoise: Really, there was nothing the matter, nothing at all. It simply couldn't reproduce the sounds on the record which I had brought him, because they were sounds that would make it vibrate and break.

Achilles: Odd, isn't it? I mean, I thought it was a Perfect phonograph. That's what the salesman had told him, after all.

Tortoise: Surely, Achilles, you don't believe everything that salesmen tell you! Are you as naive as the Crab was?

Achilles: The Crab was naiver by far! I know that salesmen are notorious prevaricators. I wasn't born yesterday!

Tortoise: In that case, maybe you can imagine that this particular salesman had somewhat exaggerated the quality of the Crab's piece of equipment ... perhaps it was indeed less than Perfect, and could not reproduce every possible sound.

Achilles: Perhaps that is an explanation. But there's no explanation for the amazing coincidence that your record had those very sounds on it ...

Tortoise: Unless they got put there deliberately. You see, before returning the Crab's visit, I went to the store where the Crab had bought his machine, and inquired as to the make. Having ascertained that, I sent off to the manufacturers for a description of its design. After receiving that by return mail, I analyzed the entire construction of the phonograph and discovered a certain set of sounds which, if they were produced anywhere in the vicinity, would set the device to shaking and eventually to falling apart.

Achilles: Nasty fellow! You needn't spell out for me the last details: that you recorded those sounds yourself, and offered the dastardly item as a gift ...

Tortoise: Clever devil! You jumped ahead of the story! But that wasn't t end of the adventure, by any means, for the Crab did r believe that his record player was at fault. He was quite stubborn. So he went out and bought a new record player, this o even more expensive, and this time the salesman promised give him double his money back in case the Crab found a soul which it could not reproduce exactly. So the Crab told r excitedly about his new model, and I promised to come over and see it.

Achilles: Tell me if I'm wrong-I bet that before you did so, you on again wrote the manufacturer, and composed and recorded new song called "I Cannot Be Played on Record Player based on the construction of the new model.

Tortoise: Utterly brilliant deduction, Achilles. You've quite got the spirit.

Achilles: So what happened this time?

Tortoise: As you might expect, precisely the same thing. The phonograph fell into innumerable pieces, and the record was shattered. Achilles: Consequently, the Crab finally became convinced that there could be no such thing as a Perfect record player.

Tortoise: Rather surprisingly, that's not quite what happened. He was sure that the next model up would fill the bill, and having twice the money, h e--

Achilles: Oho-I have an idea! He could have easily outwitted you, I obtaining a LOW-fidelity phonograph-one that was not capable of reproducing the sounds which would destroy it. In that way, he would avoid your trick.

Tortoise: Surely, but that would defeat the-original purpose-namely, to have a phonograph which could reproduce any sound whatsoever, even its own self-breaking sound, which is of coup impossible.

Achilles: That's true. I see the dilemma now. If any record player-si

Record Player X-is sufficiently high-fidelity, then when attempts to play the song "I Cannot Be Played on Record Player X", it will create just those vibrations which will cause to break. .. So it fails to be Perfect. And yet, the only way to g, around that trickery, namely for Record Player X to be c lower fidelity, even more directly ensures that it is not Perfect It seems that every record player is vulnerable to one or the other of these frailties, and hence all record players are defective.

Tortoise: I don't see why you call them "defective". It is simply an inherent fact about record players that they can't do all that you might wish them to be able to do. But if there is a defect anywhere, is not in THEM, but in your expectations of what they should b able to do! And the Crab was just full of such unrealistic expectations.

Achilles: Compassion for the Crab overwhelms me. High fidelity or low fidelity, he loses either way.

Tortoise: And so, our little game went on like *_his for a few more rounds, and eventually our friend tried to become very smart. He got wind of the principle upon which I was basing my own records, and decided to try to outfox me. He wrote to the phonograph makers, and described a device of his own invention, which they built to specification. He called it "Record Player Omega". It was considerably more sophisticated than an ordinary record player.

Achilles: Let me guess how: Did it have no of cotton? Or

Tortoise: Let me tell you, instead. That will save some time. In the first place, Record Player Omega incorporated a television camera whose purpose it was to scan any record before playing it. This camera was hooked up to a small built-in computer, which would determine exactly the nature of the sounds, by looking at the groove-patterns.

Achilles: Yes, so far so good. But what could Record Player Omega do with this information?

Tortoise: By elaborate calculations, its little computer figured out what effects the sounds would have upon its phonograph. If it deduced that the sounds were such that they would cause the machine in its present configuration to break, then it did something very clever. Old Omega contained a device which could disassemble large parts of its phonograph subunit, and rebuild them in new ways, so that it could, in effect, change its own structure. If the sounds were "dangerous", a new configuration was chosen, one to which the sounds would pose no threat, and this new configuration would then be built by the rebuilding subunit, under direction of the little computer. Only after this rebuilding operation would Record Player Omega attempt to play the record.

Achilles: Aha! That must have spelled the end of your tricks. I bet you were a little disappointed.

Tortoise: Curious that you should think so ... I don't suppose that you know Godel's Incompleteness Theorem backwards and forwards, do you?

Achilles: Know WHOSE Theorem backwards and forwards? I've

heard of anything that sounds like that. I'm sure it's fascinating, but I'd rather hear more about "music to break records by". It's an amusing little story. Actually, I guess I can fill in the end. Obviously, there was no point in going on, and so you sheepishly admitted defeat, and that was that. Isn't that exactly it?

Tortoise: What! It's almost midnight! I'm afraid it's my bedtime. I'd love to talk some more, but really I am growing quite sleepy.

Achilles: As am 1. Well, 1 u be on my way. (*As he reaches the door, he suddenly stops, and turns around*.) Oh, how silly of me! I almost forgo brought you a little present. Here. (Hands *the Tortoise a small neatly wrapped package*.)

Tortoise: Really, you shouldn't have! Why, thank you very much indeed think I'll open it now. (*Eagerly tears open the package, and ins discovers a glass goblet*.) Oh, what an exquisite goblet! Did y know that I am quite an aficionado for, of all things, gl goblets?

Achilles: Didn't have the foggiest. What an agreeable coincidence!

Tortoise: Say, if you can keep a secret, I'll let you in on something: I trying to find a Perfect goblet: one having no defects of a sort in its shape. Wouldn't it be something if this goblet-h call it "G"-were the one? Tell me, where did you come across Goblet G?

Achilles: Sorry, but that's MY little secret. But you might like to know w its maker is.

Tortoise: Pray tell, who is it?

Achilles: Ever hear of the famous glassblower Johann Sebastian Bach? Well, he wasn't exactly famous for glassblowing-but he dabbled at the art as a hobby, though hardly a soul knows it-a: this goblet is the last piece he blew.

Tortoise: Literally his last one? My gracious. If it truly was made by Bach its value is inestimable. But how are you sure of its maker

Achilles: Look at the inscription on the inside-do you see where tletters `B', `A', `C', `H' have been etched?

Tortoise: Sure enough! What an extraordinary thing. (Ge*ntly sets Goblet G down on a shelf.*) By the way, did you know that each of the four letters in\Bach's name is the name of a musical note?

Achilles:' tisn't possible, is it? After all, musical notes only go from ‘A’ through `G'.

Tortoise: Just so; in most countries, that's the case. But in Germany, Bach’s own homeland, the convention has always been similar, except that what we call `B', they call `H', and what we call `B-flat', they call `B'. For instance, we talk about Bach's "Mass in B Minor whereas they talk about his "H-moll Messe". Is that clear?

Achilles: ... hmm ... I guess so. It's a little confusing: H is B, and B B-flat. I suppose his name actually constitutes a melody, then

Tortoise: Strange but true. In fact, he worked that melody subtly into or of his most elaborate musical pieces-namely, the final *Contrapunctus* in his *Art of the Fugue*. It was the last fugue Bach ever wrote. When I heard it for the first time, I had no idea how would end. Suddenly, without warning, it broke off. And the ... dead silence. I realized immediately that was where Bach died. It is an indescribably sad moment, and the effect it had o me was-shattering. In any case, B-A-C-H is the last theme c that fugue. It is hidden inside the piece. Bach didn't point it out

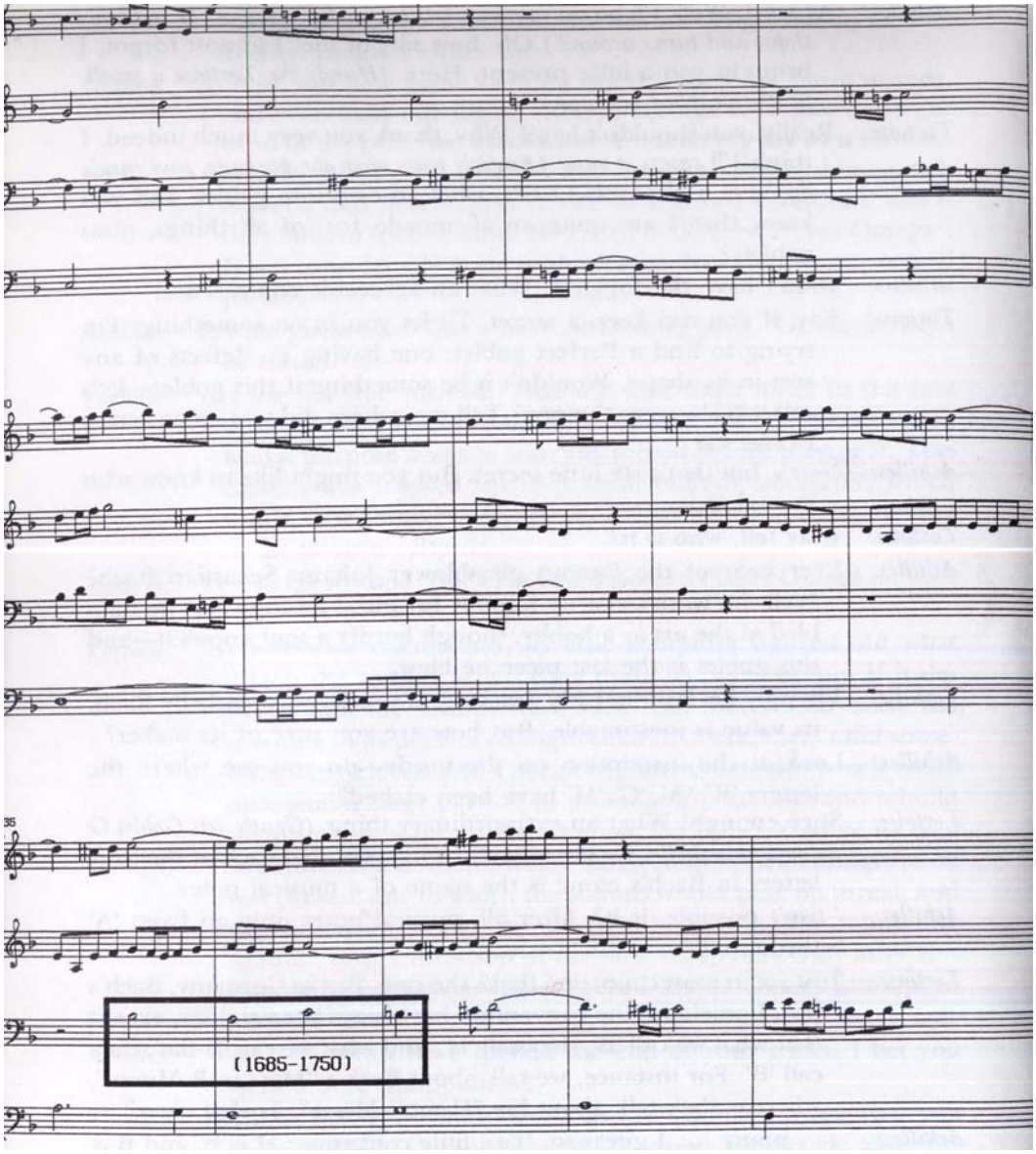

FIGURE 19. The last page of Bach's Art of the Fugue. In the original manuscript, in the handwriting of Bach's son Carl Philipp Emanuel, is written: "N.B. In the course of this fugue, at the point where the name B.A.C.H. was brought in as countersubject, the composer died." (B-A-C-H in box.) I have let this final page of Bach's last fugue serve as an epitaph.
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Explicitly, but if you know about it, you can find it without much trouble. Ah, me-there are so many clever ways of hiding things in music .. .

Achilles: . . or in poems. Poets used to do very similar things, you know (though it's rather out of style these days). For instance, Lewis Carroll often hid words and names in the first letters (or characters) of the successive lines in poems he wrote. Poems which conceal messages that way are called "acrostics".

Tortoise: Bach, too, occasionally wrote acrostics, which isn't surprising. After all, counterpoint and acrostics, with their levels of hidden meaning, have quite a bit in common. Most acrostics, however, have only one hidden level-but there is no reason that one couldn't make a double-decker-an acrostic on top of an acrostic. Or one could make a "contracrostic"-where the initial letters, taken in reverse order, form a message. Heavens! There's no end to the possibilities inherent in the form. Moreover, it's not limited to poets; anyone could write acrostics-even a dialogician.

Achilles: A dial-a-logician? That's a new one on me.

Tortoise: Correction: I said "dialogician", by which I meant a writer of dialogues. Hmm ... something just occurred to me. In the unlikely event that a dialogician should write a contrapuntal acrostic in homage to J. S. Bach, do you suppose it would be more proper for him to acrostically embed his OWN name-or that of Bach? Oh, well, why worry about such frivolous matters? Anybody who wanted to write such a piece could make up his own mind. Now getting back to Bach's melodic name, did you know that the melody B-A-C-H, if played upside down and backwards, is exactly the same as the original?

Achilles: How can anything be played upside down? Backwards, I can see-you get H-C-A-B-but upside down? You must be pulling my leg.

Tortoise: ' pon my word, you're quite a skeptic, aren't you? Well, I guess I'll have to give you a demonstration. Let me just go and fetch my fiddle- (Walks into the next room, and returns in a jiffy with an ancient-looking violin.) -and play it for you forwards and backwards and every which way. Let's see, now ... (Places his copy of the Art of the Fugue on his music stand and opens it to the last page.) ... here's the last Contrapunctus, and here's the last theme ...

The Tortoise begins to play: B-A-C- - but as he bows the final H, suddenly, without warning, a shattering sound rudely interrupts his performance. Both he and Achilles spin around, just in time to catch a glimpse of myriad fragments of glass tinkling to the floor from the shelf where Goblet G had stood, only moments before. And then ... dead silence.

Chapter IV

Consistency, Completeness, and Geometry

Implicit and Explicit Meaning

IN CHAPTER II, we saw how meaning-at least in the relatively simple context of formal systems-arises when there is an isomorphism between rule-governed symbols, and things in the real world. The more complex the isomorphism, in general, the more "equipment"-both hardware and software-is required to extract the meaning from the symbols. If an isomorphism is very simple (or very familiar), we are tempted to say that the meaning which it allows us to see is explicit. We see the meaning without seeing the isomorphism. The most blatant example is human language, where people often attribute meaning to words in themselves, without being in the slightest aware of the very complex "isomorphism" that imbues them with meanings. This is an easy enough error to make. It attributes all the meaning to the object (the word), rather than to the link between that object and the real world. You might compare it to the naive belief that noise is a necessary side effect of any collision of two objects. This is a false belief; if two objects collide in a vacuum, there will be no noise at all. Here again, the error stems from attributing the noise exclusively to the collision, and not recognizing the role of the medium, which carries it from the objects to the ear.

Above, I used the word "isomorphism" in quotes to indicate that it must be taken with a grain of salt. The symbolic processes which underlie the understanding of human language are so much more complex than the symbolic processes in typical formal systems, that, if we want to continue thinking of meaning as mediated by isomorphisms, we shall have to adopt a far more flexible conception of what isomorphisms can be than we have up till now. In my opinion, in fact, the key element in answering the question "What is consciousness?" will be the unraveling of the nature of the "isomorphism" which underlies meaning.

Explicit Meaning of the *Contracrostipunctus*

All this is by way of preparation for a discussion of the *Contracrostipunctus*-a study in levels of meaning. The Dialogue has both explicit and implicit meanings. Its most explicit meaning is simply the story